Enhancing Critical Thinking in Education by Means of a Socratic Chatbot

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Abstract. While large language models (LLMs) are increasingly playing a pivotal role in education by providing instantaneous, adaptive responses, their potential to promote critical thinking remains understudied. In this paper, we fill such a gap and present an innovative educational chatbot designed to foster critical thinking through Socratic questioning. Unlike traditional intelligent tutoring systems, including educational chatbots, that tend to offer direct answers, the proposed Socratic tutor encourages students to explore various perspectives and engage in self-reflection by posing structured, thought-provoking questions. Our Socratic questioning is implemented by fine and prompt-tuning the open-source pretrained LLM with a specialized dataset that stimulates critical thinking and offers multiple viewpoints. In an effort to democratize access and to protect the students' privacy, the proposed tutor is based on small LLMs (Llama2 7B and 13B-parameter models) that are able to run locally on off-the-shelf hardware. We validate our approach in a battery of experiments consisting of interactions between a simulated student and the chatbot to evaluate its effectiveness in enhancing critical thinking skills. Results indicate that the Socratic tutor supports the development of reflection and critical thinking significantly better than standard chatbots. Our approach opens the door for improving educational outcomes by cultivating active learning and encouraging intellectual autonomy.

Keywords: Critical Thinking · Socratic Chatbot · Education

1 Introduction

Recent advancements in generative artificial intelligence (AI) methods, combined with an almost universal availability of generative AI tools, have expanded the

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potential to assist people in developing new ways to think, communicate, learn, and solve problems creatively [18]. The integration of large language models (LLMs) in education through interactive chatbots has emerged as a transformative element in the evolving educational technology landscape, opening new avenues for learning and pedagogy [24].

While recent work illustrates the potential positive impact of chatbots in learning [3], their capabilities to support essential aspects of human learning, such as critical thinking and self-regulated learning, are limited [25]. However, addressing such limitations is of paramount importance, as these aspects are widely regarded as fundamental pillars in human learning yet inadequately accomplished with current technologies [26].

In this paper, we focus on tackling these limitations. We present a novel Socratic chatbot designed to align with pedagogical principles by promoting critical thinking, purposeful learning and self-efficacy. We propose the use of Socratic dialogues to enhance learning experiences by having the chatbot ask questions rather than provide immediate answers. Our approach combines the traditional elements of Socratic questioning with the innovative capabilities of AI, aiming to create a personalized, responsive and enjoyable learning experience. Furthermore, we are interested in exploring the capabilities of small, open-source LLMs running locally on the learners' laptops to act as Socratic tutors. Locally run LLMs provide more privacy and contribute to the democratization of education as they do not need a constant internet connection benefiting students in rural or underserved areas.

This paper is organized as follows: the most relevant related literature is presented in Sect. 2, establishing the background and context for our research. Section 3 describes our approach to develop a Socratic chatbot, consisting of (1) fine-tuning an open-source pretrained large language model on a dataset designed to enhance critical thinking skills and (2) prompt-tuning it to create conversations according to the Socratic method. Our experimental evaluation and comparison of the Socratic tutor with two baselines is described in Sect. 4, followed by the result in Sect. 5, a discussion in Sect. 6 and a conclusion and outline of future lines of research in Sect. 7. Furthermore, a discussion about the ethical implications and challenges is provided in the supplementary material.

2 Related Work

In this section, we first provide an overview of the most relevant previous works in critical thinking and Socratic questioning, both from an educational and technological perspective.

2.1 Educational Chatbots

For the past fifteen years, scholars have explored several facets of integrating chatbots into educational settings, examining their effectiveness, benefits, and challenges [34]. Early work investigated basic interactions and information

retrieval tasks [23]. As AI methods improved, studies explored the role of chatbots in personalized learning [37], student engagement [14], and language learning [13]. Recently, the use of LLM-based chatbots in education has gained significant research attention and societal adoption with the emergence of numerous EdTech startups devoted to this topic [35]. Yet, there is a lack of educational chatbots to support critical thinking and self-regulated learning.

2.2 Critical Thinking

Definition in education. Critical thinking (CT) is a complex cognitive process involving the evaluation, analysis, and synthesis of information to form reasoned judgments [2,5,33]. Essential across all educational levels and disciplines, CT empowers students to think independently, solve problems, and make informed decisions. Several strategies are effective in promoting CT skills [2], such as: (1) engaging students in structured dialogues to explore various perspectives; (2) exposing students to real-world scenarios requires them to actively apply their CT skills; and (3) providing personalized mentoring to support the development of CT capabilities.

However, assessing CT skills presents difficulties due to a lack of a standardized evaluation framework, complicating efforts to further integrate and enhance CT training in educational environments [5].

Chatbots for Critical Thinking. Incorporating chatbot technology into educational settings offers substantial benefits for enhancing CT [32]. Indeed, chatbots can effectively employ the three previously mentioned strategies by facilitating guided dialogues focused on specific, concrete problems. Although research in this area is still emerging, some progress has been reported in the past two years. For example, the AI-IESLS chatbot enhances the students' learning experiences and critical thinking in higher education through the use of concept mapping and probability distribution analysis [29]. Moreover, a chatbot designed according to cognitive styles has been proven effective in advancing science concepts and critical thinking skills among preparatory school students [8]. Finally, the online education nonprofit Khan Academy has recently developed Khanmigo, an AI-powered tutor and teaching assistant hand-crafted by their learning experts. It aims to promote critical thinking by guiding students through interactive problem-solving[1].

2.3 Socratic Questioning

Definition in Education. Named after the classical Greek philosopher Socrates, Socratic questioning is a structured way of asking questions to explore complex ideas, reveal underlying assumptions, and discern what is known from what is not [33].

[1] https://www.khanmigo.ai.

Socratic questioning is part of the scaffolding theory of learning, a process in which a learner achieves a goal through guided efforts [40]. Students are prompted with questions that guide them towards the solution by solving problems beyond their zone of proximal development [39].

Its primary purpose in education is to test and deepen the students' understanding, encourage them to ask profound questions, help them think critically about complex topics, and foster an understanding of different viewpoints [33]. Socratic questioning has been shown to promote critical thinking, improve the student's understanding and encourage a culture of inquiry and reflection, essential for reasoning through complex issues and understanding diverse perspectives [11,19,41].

However, the effectiveness of the Socratic method varies among learners because it depends on their ability to abstract and generalize knowledge [15]. Thus, this method should be adapted to individual differences in cognitive processing, prior knowledge, or learning styles [33].

Chatbots for Socratic Questioning. Chatbots can support the Socratic method of questioning by engaging users in a guided dialogue, consisting of probing, open-ended questions that encourage critical thinking and deep reflection [5,36].

To the best of our knowledge, we are unaware of any fully implemented chatbot designed to leverage the Socratic questioning method to enhance critical thinking in human learning. Nonetheless, recent advancements have applied machine learning techniques (like zero-shot prompt tuning and reinforcement learning) to automatically generate relevant Socratic questions in diverse contexts: math word problems [36], debugging code [4], physics [16], and cognitive behavioral therapy [22].

Furthermore, Ang et al. [6] introduced *SocratiQ*, a dataset consisting of approximately 110,000 Socratic question-context pairs curated from the subreddit *r/changemyview*[2]. The authors employed a transformer-based encoder to classify each question-context pair into specific Socratic question types as detailed in Table 2, in the supplementary material. Subsequently, for each identified type of Socratic question, a decoder-like language model was trained to generate new questions corresponding to that specific type from a given context. Both automated and human evaluation studies demonstrated that these models successfully produce type-sensitive, relevant, and human-like Socratic questions.

Filling the existing gap in the literature, our Socratic tutor's core design principle is the use of Socratic questioning to guide the student's learning process, changing the focus from providing answers to formulating questions that would support the learner's critical thinking. In our approach, we leverage the *SocratiQ* annotated dataset of question-context pairs to fine-tune an open-source, pretrained LLM to generate specific types of Socratic questions.

2.4 Contributions

Given the existing literature, the contributions of our paper are five-fold:

[2] https://www.reddit.com/r/changemyview/.

1. We develop an LLM model focused on critical thinking by fine and prompt-tuning open-source, pretrained small LLMs (Llama2 7B- and 13B-parameter models) to create a Socratic tutor that poses a variety of Socratic questions in a computationally efficient manner;
2. We propose an approach for generating tutoring dialogues by means of an LLM-based dual-agent (tutor-learner) setup;
3. We introduce an extrinsic evaluation method that assesses the quality of the questions by comparing the learner's responses to the ideal responses based on the problem description;
4. We implement an automatic evaluation mechanism using an LLM that is designed to replicate human evaluative behaviors; and
5. We validate our approach on a Theory of Knowledge task and compare it with two baselines.

3 Socratic Tutor Implementation

In this section, we describe the fine and prompt-tuning approach that we adopted to build our Socratic tutor. We experiment with two open-source, pretrained large language models: Llama2 Instruct with 7B parameters and Llama2 Instruct with 13B parameters [38] which are fine-tuned on Google Colab with a L4 GPU. The fine-tuned models are then run on an Apple M1 Pro laptop with 32 GB RAM. This approach of using small models on a local computer is chosen to enable students to operate the chatbot independently on their personal computers, with the aim of democratizing education and protecting privacy. A key challenge in this regard lies in optimizing the chatbot's performance to ensure it operates efficiently on local machines without excessive resource consumption.

3.1 Socratic Fine-Tuning

To fine-tune the open-source, pretrained LLMs we applied parameter-efficient fine-tuning (PEFT) techniques: Low-Rank Adaptation (LoRA) [20] and Quantized Low-Rank Adaptation (QLoRA) [12]. They offer a way to effectively adapt LLMs to specific tasks without the need to retrain or alter the entire pre-trained model. LoRA focuses on modifying a small number of additional parameters that act upon the pre-trained weights of the model. This approach is especially valuable due to its efficiency and the reduced computational overhead it introduces. Building upon LoRA, QLoRA incorporates quantization into the fine-tuning process (4-bit quantization in our case). It reduces the precision of the low-rank parameters to further decrease memory usage and computational demands.

Hyperparameters. LLMs were fine-tuned on a single GPU, using mini-batches of size 4. We adopted a learning rate of 2×10^{-4} and used the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) in 32-bit precision and a cosine-based scheduler with a step ratio for linear warmup of 0.03. A weight decay of 0.001 was applied to all weights except biases and normalization layer parameters. Regarding QLoRA, the rank of LoRA modules, r is 64, the LoRA scaling factor, α is 16 with 0.1 dropout and 4-bit quantization.

Fine-Tuning Dataset. The fine-tuning process requires a specialized dataset. In our case, we used the *SocratiQ* dataset [6] which collects contentious viewpoints and the questions challenging these viewpoints from Reddit (Subreddit channel called *subreddit r/changemyview*). This channel corresponds to an active community specifically designed as "*a place to post an opinion you acknowledge may be flawed, in an effort to understand other perspectives on the issue*". Discussions often revolve around controversial topics, such as politics, media, and culture. Members are encouraged to approach these discussions with an openness to dialogue rather than debate. Due to its foundational goal of encouraging users to "change their view", comments frequently include questions that promote introspection and reflection.

Ang et al. [6] classified the context-question pairs of this Subreddit into the Socratic categories devised by Paul and Elder [33]. The question types along with definitions and examples can be found in Table 2. These categories address intellectual standards to enhance the rigor of Socratic dialogues [33].

The fine-tuning was performed with a subset of 600 context-question types of questions from *SocratiQ* that had been annotated through the Amazon Mechanical Turk crowdsourcing platform [10]. Given a context and a specific type of question, the LLMs are fine-tuned to generate Socratic questions of this kind.

3.2 Socratic Prompt-Tuning

In addition, we prompt-tuned the fine-tuned LLMs. Prompt-tuning is a method of fine-tuning a language model on specific tasks by adjusting its input prompts without altering the underlying model. We prompt-tuned three different types of LLM-based tutors: the Socratic tutor and two baselines (described in Sect. 4.3) which correspond to a basic tutor, and a random tutor. The prompts provided to the Socratic tutor are composed of the items listed below[3] and reflected in the first row of Table 3, in the supplementary material:

- **Role.** In the first part of the prompt, we inform the LLM of its role. In the case of the Socratic tutor, it is instructed to be a philosopher who follows Socratic methods, which involves asking probing questions to encourage deep thinking and challenge assumptions.
- **Goal.** The second part contains the goal of the interaction, namely to help a student think about a question (*Theory of Knowledge or ToK question*).
- **Resources.** Then, the prompt imposes the restriction of using only the provided dialogue history (*history*) and the student's latest response (*input*). These elements serve as the context and content from which the LLM must derive its question, underscoring that the tutor should not teach or introduce new material, but instead focus on exploring the students' thoughts and facilitating their reflection, consistent with the principles of Socratic dialectic [33].

[3] We direct the reader to the supplementary material for a description of the prompt engineering performed in this paper.

– **Expected outcome.** Finally, the prompt specifies the expected outcome. The question asked should logically continue the ongoing conversation and must be designed to lead the student to explore their thoughts and ideas further rather than providing direct answers.

Table 4, in the supplementary material, provides an example of a simulated conversation with our *Socratic tutor*.

4 Automatic Evaluation

To evaluate the effectiveness of the proposed Socratic tutor in fostering critical thinking, we simulate a dialogue between our Socratic tutor and an LLM-based simulated learner whose task is to respond to a specific Theory of Knowledge (ToK, see below) question. While an empirical evaluation with human subjects would be ideal and it is planned as a next step, simulated evaluations are necessary for several reasons. First, they enable a preliminary assessment of the functionality, efficiency, and safety of the approach. Second, they allow for the creation of benchmark data for comparison against human interaction. Finally, they enable running numerous simulated conversations to gather data and refine the chatbot's behavior without the logistical challenges associated with human participants.

4.1 Domain: Theory of Knowledge (ToK)

The *Theory of Knowledge* (ToK) course is part of the International Baccalaureate (IB) Diploma Programme[4]. It aims to develop critical thinking by exploring the process of knowing rather than teaching specific knowledge[5]. The course's main goals include linking knowledge construction to academic disciplines and the real world, fostering awareness of diverse cultural perspectives, and encouraging personal reflection on beliefs and assumptions.

However, ToK courses and assessments, do not fully capture the depth of critical thinking skills intended, raising concerns about their validity [21]. To enhance its effectiveness, the course could benefit from a broader inclusion of critical thinking skills, particularly in creative and active thinking, and adjustments to its assessment strategies to better align with these skills [21]. A chatbot designed to promote critical thinking could engage and support students in diverse and complex questioning, encouraging them to explore and evaluate multiple perspectives and scenarios actively. Specifically, the final assessment for this course comprises two main components: an exhibition and an essay. The latter, limited to 1,600 words, addresses a theoretical question within ToK. It typically prompts students to analyze and discuss claims about the nature of knowledge

[4] https://www.ibo.org/.
[5] https://www.ibo.org/programmes/diploma-programme/curriculum/dp-core/theory-of-knowledge/what-is-tok/.

production. An example essay topic might require students to explore the assertion that the methods employed in generating knowledge are influenced by the intended application of that knowledge (see footnote 5). Table 7, in the supplementary material, includes examples of ToK questions from the International Baccalaureate curriculum[6].

Our approach is to use our Socratic chatbot to help a learner think about these ToK questions to help them develop and articulate logically their ideas to complete the essay.

4.2 Generation of Synthetic Conversations Using LLMs

In order to empirically evaluate the proposed Socratic tutor we generate synthetic dialogues using two LLM-based agents that engage in a conversation with each other, as has been recently done in the literature with the purpose of assisting users in building linear programming models [1]. In our case, we create an LLM-based conversational student that generates the learner's responses by prompt-tuning a small LLM (Llama2 7B-parameter model) with the following prompt: "ANSWER to the question <input> IN ONE SENTENCE. DO NOT USE COMPLEX WORDS OR IDEAS".

4.3 Baselines: Non-socratic Tutors

To evaluate the effectiveness of the proposed Socratic tutor, we benchmark it against two non-Socratic LLM-based tutoring methods: (1) A *basic tutor* which engages learners in discussions about the Theory of Knowledge questions without a specific intention to help them; and (2) a *random tutor* which produces arbitrary responses and serves as a control in our study. We create these two tutors by prompt-tuning the open-source, pretrained Llama2 7B-parameter model according to the prompts described in Table 3, in the supplementary material. As a result, we create three different tutors (Socratic, basic and random).

Additional results comparing the Socratic tutor and a baseline tutor that had not been neither fine- nor prompt-tuned can be found in the supplementary material.

We test each of these tutors on 5 different ToK questions displayed in Table 7 in the supplementary material. For each tutor and question, we perform 20 conversations of 5 turns each. Given that the topic of the conversations is cognitively demanding and thought-provoking, our Socratic chatbot should enable students to successfully respond to the task within a limited number of interactions. Thus, we set the number of turns to 5. We also include results corresponding to longer conversations with 20 turns in Fig. 6 in the supplementary material.

[6] https://ibtokessaytutor.com/ibtokessaytopics/May2023/ib-tok-essay-titles-May-2023.shtml.

4.4 Evaluation Metrics

We compare the performance of the three types of chatbots by means of an extrinsic evaluation approach [1]. We assess the quality of the questions generated by the tutors by analyzing the responses provided by the simulated learners. This evaluation operates under a key hypothesis: if the learner delivers accurate responses, this suggests that the tutor has posed effective questions that led the learner to generate such answers. Thus, we measure the effectiveness of the questions by comparing the concatenated learner's responses to a summary of the essential points that are expected in the responses to each of the 5 ToK questions. Tables 8, 9 and 10 in the supplementary material, include examples of the ToK questions used with a summary of the information that should be included in the answer to the questions.

We use standard n-gram-based metrics—BLEU [28,31], ROUGE-L [27] and METEOR [7]— to evaluate the quality of the responses provided by the learners [30]. However these commonly-used measures struggle to accurately assess paraphrases that, while different in wording, are equally valid interpretations of the reference questions. Thus, we use the BERTScore [42] to overcome these limitations. Metrics leveraging BERT-based models have demonstrated robustness in evaluating text generation tasks, including summarization and translation, by providing more nuanced assessments of textual similarity [17,42]. Finally, we propose a novel evaluation metric by means of an LLM [1] to specifically assess the level of critical thinking in the learner's response. The highest possible value of these metrics is 1. Below is a summary of the advantages and limitations of the BERTScore and LLM-based metrics. See the supplementary material for the other metrics.

- **BERTScore:** Originally, the BERTScore [42] leveraged pre-trained contextual embeddings from BERT to evaluate the similarity between words in candidate and reference sentences using cosine similarity. This metric effectively aligns with human judgment, demonstrating strong correlation in both sentence-level and system-level evaluations [42].
  However, *DeBERTa* [17] enhances both the BERT and RoBERTa architectures by incorporating disentangled attention mechanisms and an improved mask decoder. With 80 GB of training data, DeBERTa surpasses both BERT and RoBERTa on the majority of natural language understanding tasks, thereby demonstrating superior performance across a broad range of evaluations.
- **LLM-score:** Previous metrics primarily assessed the similarity between the learner's response and a summary of potential answers to the formulated question. To more directly evaluate critical thinking, we introduce a score generated by a pretrained Llama2 7B-parameter LLM [38] which is specifically prompted to assess the level of critical thinking in the learner's response. The evaluation prompt is composed of the following elements: (1) *role*, provided in the first part of the prompt where it specifies that the LLM should act as an AI evaluator specialized in assessing critical thinking in discourse; (2)

definition of critical thinking as per [9]; (3) *goal*, which is expressed as the evaluation of the use of critical thinking in the learner's response to a specific ToK question; and (4) *outcome*, which is a score ranging from 0 to 5, reflecting the degree of critical thinking evident in the responses. This score should be accompanied by an explanation, employing the chain-of-thought prompting technique. Ultimately, the LLM-score is normalized to a scale between 0 and 1 to be comparable to the rest of scores. Table 11, in the supplementary material, provides a summary of the prompt used to generate the LLM-score.

The code to create the LLM-based tutors and learner, and to run the experiments described in this paper can be found here: https://github.com/lucilefavero/ECAI_2024.

5 Results

Tables 4, 5 and 6, in the supplementary material, provide examples of simulated conversations with the *Socratic tutor*, the *basic tutor*, and the *random tutor*, respectively, all generated with the Llama2 7B-parameter LLM. We also create another Socratic tutor using the Llama2 13B-parameter LLM to shed light on the impact of increasing the complexity of the model to achieve the task. The performance of these tutors —quantified using the BLEU, ROUGE-L, METEOR, BERTScore and LLM score– are presented in Table 1, corresponding to the average values of 20 conversations of 5 turns for each 5 of the ToK questions. The differences in performance between the Socratic tutors (indicated with bold font) and the non-Socratic tutors are statistically significant (t-test with p-value < 0.001).

Table 1. Performance of the different types of tutors, averaged over 20 conversations of 5 turns each for each of the 5 ToK questions. The differences between the Socratic tutors (highlighted in bold) and the non-Socratic tutors are statistically significant (t-test, p-value < 0.001). The differences in performance between the two Socratic tutors are not statistically significant.

Tutor type	BLEU	ROUGE-L	METEOR	BERTScore	LLM Score
Socratic Llama2 13B	**3.65**	**0.157**	**0.226**	**0.569**	**0.696**
Socratic Llama2 7B	**3.42**	**0.162**	**0.216**	**0.576**	**0.670**
Basic Llama2 7B	0.494	0.120	0.092	0.535	0.582
Random Llama2 7B	0.210	0.091	0.063	0.444	0.312

Figure 1, Fig. 2, and Fig. 3 depict the performance (METEOR, BERT and LLM scores) of the different types of tutors across 5 turns for 20 conversations for each 5 of the ToK questions. Figure 4 and Fig. 5, in the supplementary material display the performance of the BLEU and ROUGE-L scores. As seen in

the figures, for all metrics and the Socratic and basic tutors, the scores increase with the number of turns. Furthermore, the *Socratic tutor* significantly outperforms the other two tutors, demonstrating a more pronounced improvement as the conversation evolves. While the performance of the Socratic tutor implemented with the Llama2 13B-parameter model is superior to that of the Socratic tutor based on the Llama2 7B-parameter model, the differences in performance between these two tutors are not statistically significant.

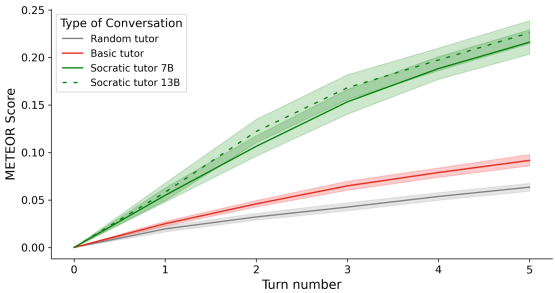


Fig. 1. METEOR scores of the different types of tutors, averaged over 20 conversations of 5 turns each and for each of the 5 ToK questions. The differences between the Socratic tutor and the non-Socratic tutors are statistically significant (t-test, p-value < 0.001). No statistically significant difference is observed in the performance of both Socratic tutors.

6 Discussion

From our study, we draw several implications.

First, our results clearly demonstrate that the Socratic tutor outperformed both the basic and random tutors in assisting students with ToK questions. After only 5 turns of interaction, students engaging with the Socratic tutor showed a significant improvement in their critical thinking abilities compared to those interacting with the baseline chatbots. This superior performance indicates the effectiveness of employing Socratic questioning techniques in guiding students through the process of critical thinking.

Second, by developing educational chatbots based on small LLMs that can be run on personal computers, our methodology lowers the barriers of access to educational resources, democratizing education across different demographics and regions. Thus, our approach contributes to greater educational equity by enabling individuals to independently access and leverage advanced technological tools for learning, regardless of their socio-economic status or geographical location.

Third, as a preliminary evaluation, we assess the effectiveness of our method through dialogues with a simulated learner powered by an LLM. Our experimental methodology contributes to the growing use of LLMs to simulate human

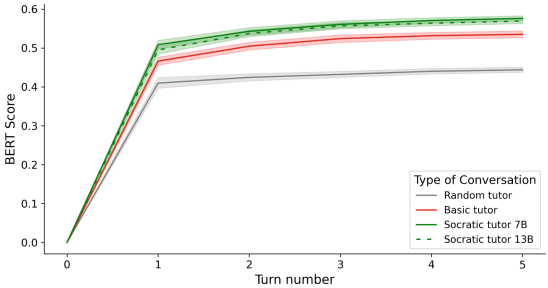


Fig. 2. BERT scores of the different types of tutors, averaged over 20 conversations of 5 turns each and for each of the 5 ToK questions. The differences between the Socratic tutors and the non-Socratic tutors are statistically significant (t-test, p-value < 0.001). No statistically significant difference is observed in the performance of both Socratic tutors.

interactions in data generation tasks. Moving forward, we plan to conduct case studies involving actual students to further validate our method.

Fourth, despite the promising results obtained by leveraging fine and prompt-tuned pre-trained small LLMs for this task, we encountered challenges in prompt-tuning. Designing prompts that effectively elicit critical thinking and guide students through the reasoning process proved to be a complex task. Occasionally, our model would generate dialogues where the tutor both posed questions and provided answers, or indirectly guided the conversation by stating phrases such as *"If I were a Socratic tutor, I would ask..." or "My response to the student is..."*. After extensive empirical work with different types of prompts, we discovered that removing the memory of the dialogue from the learner's prompt substantially enhanced the coherence and logical flow of the conversations, leading to higher-quality interactions.

Fifth, surprisingly, the performance of the Socratic tutor based on the smallest LLM is at par with that of the Socratic tutor built from an LLM with almost twice the number of parameters. These results illustrate the power of even small LLMs to provide educational value even in contexts where resource constraints or computational limitations may be a concern. We leave to future work a more in-depth exploration of how to improve the performance of the Socratic tutor based on the Llama3 8B-parameter model or models of similar complexity.

Sixth, our study's findings have significant implications for education, particularly in the context of online learning and intelligent tutoring systems. Integrating Socratic chatbots into educational platforms can provide students with personalized, interactive, and scaffolded learning experiences that promote the development of critical thinking skills. With our Socratic tutor, educators can augment traditional instruction methods and enhance students' ability to analyze complex issues, evaluate evidence, and construct well-reasoned arguments.

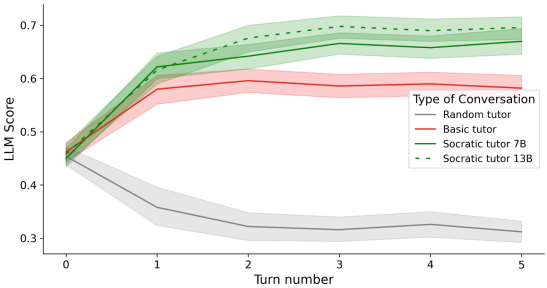


Fig. 3. Critical thinking scores obtained by means of the LLM-score for the different types of tutors. The plots display the average values over 20 conversations of 5 turns for each of the 5 ToK questions. The differences between the Socratic tutor and the non-Socratic tutors are statistically significant (t-test, p-value < 0.001). No statistically significant difference is observed in the performance of both Socratic tutors.

7 Conclusion

Our study contributes to the development of AI-driven tutoring systems by introducing a fine- and prompt-tuned model specifically aimed at generating Socratic questions to enhance critical thinking skills. Our approach enables the creation of a Socratic tutor that surpasses the baselines in delivering effective tutoring conversations to foster critical thinking. The comparison highlights the potential of tailored AI approaches in educational settings.

Looking ahead, we propose several directions for future research. First, further research is needed to continue improving the effectiveness of our Socratic tutor in educational contexts, including the study of sentiment analysis, semantic understanding and reinforcement learning with human feedback to enhance the chatbot's ability to interpret and respond to students' queries more accurately. Second, our approach should be evaluated with human students to measure its effectiveness in real educational settings and gather qualitative feedback on its tutoring capabilities. Third, there is a need to propose more refined evaluation methods that are able to adequately assess critical thinking in the context of automated tutoring conversations. Additionally, investigating the long-term impact of Socratic chatbot interventions on students' critical thinking skills and academic performance would provide valuable insights into the efficacy of this approach. Furthermore, developing a learner model that adapts to individual student responses and learning progress could greatly enhance the personalization and effectiveness of the Socratic tutor. These steps will be crucial in advancing the field of AI-driven educational tools and in creating more effective, responsive and human-centered learning environments.

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